3 THE NUCLEAR MIRAGE

I wanted nuclear to work.

Not just because it would be the answer to our civilization-ending lack of fossil fuels. There is something genuinely wonderful about harnessing the power of the atom. When I first learned about nuclear energy as a teenager, it seemed like the obvious solution. Vast quantities of low-carbon electricity from improbably small amounts of matter. If this book were written by my childhood self, this would be a jubilant chapter about building reactors everywhere.

The promise seemed so real. In 1954, Lewis Strauss, chairman of the U.S. Atomic Energy Commission, predicted that nuclear power would become "too cheap to meter." (NRC, 2021) Clean, dense, powerful, and seemingly limitless. I believed in that promise for a long time.

But I cannot, in good conscience, present nuclear as a solution in this book. Not because it isn't impressive. It is. Not because it isn't low-carbon. It is. Not because the waste problem is unsolvable. It is. And not because it isn't safer than fossil fuel. It's all these things. But because of four problems that I call the four horsemen of my nuclear disillusionment. Taken together, they unfortunately disqualify it from the future we need to build.

3.1 How Nuclear Energy Works: Just Another Way to Boil Water

Before we examine those four horsemen, it helps to understand what nuclear power actually is. In essence it's a new way to boil water. That is not meant to be derogatory. Boiling water is how we currently produce most of our electricity. We boil water, which produces steam, which turns a turbine, which generates electricity.

In a fossil fuelled power plant the heat for boiling water comes from combining carbon atoms and oxygen atoms which produces CO2 and heat. We call this process a fire. It's a chemical reaction that produced our energy for almost a million years.

In a nuclear power plant the heat comes from nuclear fuel. It works by splitting atoms in a process we call nuclear fission. Usually the atoms we split are uranium atoms so let's stick to that in our short explanation.

As you might know from school, an atom consists of a core with electrons swarming around it. The core is responsible for the atom weight. The core is called the nucleus and consists of two types of particles. Protons that are positively charged, and neutrons that are just as heavy but are not charged.

Each atom is named for the amount of protons in its nucleus. Uranium is a super heavy atom with 92 protons in its nucleus. But you can find it in two flavours that have different amounts of neutrons. These flavours are called isotopes. Most uranium occurring in nature has 146 neutrons in its nucleus which, together with the mandatory 92 protons, gives it a total weight of 238. That's why we call this isotope U-238.

But we are mostly interested in the lighter isotope. It's more unstable and hot-headed cousin, U-235. U-235 plays hard to get and in nature it is always found in small quantities together with a lot of U-238 that it hides behind.

You might have heard about *enrichment*. Enrichment is a process that Iran might be using to prepare atom bombs. Enrichment is also the process we use for creating nuclear fuel. In essence it just means that we take naturally occurring uranium with very little U-235 and we spin it in a centrifuge until the heavier U-238 goes one way and the lighter U-235. The end result is some uranium that is richer in U-235.

Now the fun starts. When you hit U-235 with a neutron, it temporarily becomes U-236 but that's so unstable that it immediately splits into three loose neutrons plus other atoms like krypton and barium. That's why it's called nuclear fission. From the latin fissio meaning splitting and related to the word *fissure* meaning a crack or split.

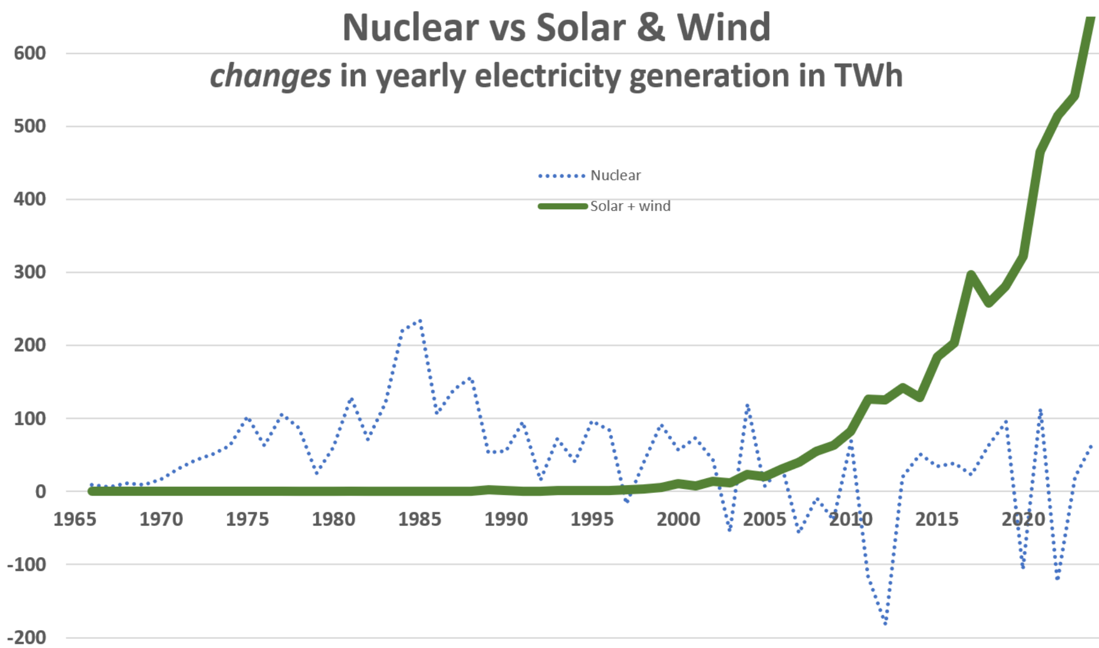
This is a big deal because the result of this reaction is a tiny bit lighter than U-236. And maybe you've heard of Einstein and the formula *energy is mass times the speed of light squared?* It doesn't matter if you haven't. Bottom line is that even that tiny amount of mass that went missing, goes a long way to producing heat.

On top of that we also got three loose neutrons. And we had all these unstable U-235 types that fall apart as soon as you hit them by a neutron. See where this is going? A chain reaction.

In a bomb the chain reaction is unchecked. In a nuclear power plants there are movable rods with material that absorbs neutrons to control the reaction. And that's how it boils water.

3.2 Yearly Growth Is Trending Down Since 1985

Nuclear power had its moment in the 1980s but then its popularity waned and it never recovered. While experts debate nuclear in conferences and op-eds, the data tells a clear story. Nuclear market share started dropping in 1996, but if you look deeper you see that the growth per year actually peaked in 1985. After that the yearly growth became less and less. So nuclear lost its momentum long before solar and wind began their exponential rise. (Carbon Brief, 2016) (WNISR, 2025)



By 1996, fossil fuels had reclaimed market share, and nuclear began its long decline. Today, solar and wind are taking that share instead.

3.3 From "Too Cheap to Meter" to "Too Expensive to Matter"

If nuclear power were merely expensive, we could debate whether the premium was worth paying. But nuclear is three times the cost of alternatives. That's a price premium no investor can overlook.

Lazard's 2024 LCOE analysis shows new nuclear at $0.142 to $0.222 per kilowatt-hour. (Lazard, 2024) More expensive than gas combined cycle. Around three times as expensive as both wind and utility-scale solar. Fraunhofer and CSIRO put nuclear's costs even higher. (Fraunhofer, 2024) (Graham, 2024) Yes, you must include storage for when the sun doesn't shine and the wind doesn't blow. But in a well-designed system, that makes surprisingly little difference, as we'll see later.

Then there are the cost overruns. Nuclear projects consistently promise one price and deliver another, usually two to three times higher. This is not occasional misfortune but a persistent pattern that makes investors wary and they make sure the risk shifts to taxpayers before they agree to build. (Sovacool, 2014) (Lovering, 2016)

Nuclear has also achieved something remarkable: negative learning. Most technologies get cheaper as production increases. We call this a learning curve, and it is so important that the next chapter is devoted entirely to understanding it. Solar panels, wind turbines, batteries, power electronics: all follow learning curves. The IEA and NEA use a method called NOAK (nth of a kind) that confidently predicts this will happen to nuclear too. They have been confidently predicting this for decades now. Unfortunately, empirical data shows the opposite is true. Newer plants cost more, not less. (Grubler, 2010) (Koomey, 2017) There are occasional exceptions, and in China everything is cheaper, including nuclear power plants, and South Korea has also achieved cost stability. Maybe someday nuclear will buck the trend elsewhere. (Lovering, 2016) (Liu, 2025) But so far, all the empirical evidence—as opposed to the aspirational predictions—shows nuclear has been on a negative learning curve.

Finally, there are hidden costs. About 1.5% of nuclear reactors that are built have experienced core meltdowns. If their number doesn't grow this would statistically mean a major accident somewhere in the world roughly every 10–20 years. (Wheatley, 2017) Chernobyl and Fukushima together tally close to $1 trillion in direct financial damages, with estimates ballooning when ongoing and indirect costs are included. (Samet, 2016) Spread that over total historical nuclear generation and you get about one cent per kilowatt-hour and a lot of uncertainty. Add waste storage—uncertain, multigenerational, and expensive—and the economics become difficult to justify.

There is also the cost of being slow. Nuclear plants take a decade longer to build than renewables. (Jacobson, 2020) During that decade, fossil fuels continue emitting greenhouse gases and killing people through air pollution. If we are honest about cost-benefit analysis, those deaths and emissions belong on nuclear's ledger too.

3.4 More Nuclear Energy Means More Nuclear Weapons

That nuclear power plants enable nuclear weapons is the elephant tapdancing in the room that most polite society prefers to tiptoe around. Preferably while humming. The link between nuclear power and nuclear weapons remains debated, but empirical evidence suggests civilian nuclear cooperation increases proliferation risks. (Fuhrmann, 2009)

Today we have about 415 nuclear reactors in 31 countries. (IAEA, 2024) In total, about 30 countries have sought nuclear weapons, and ten are known to have succeeded. (ISIS, 2004) Of these 10, seven have a civilian nuclear program. If nuclear were to replace the energy from fossil fuels worldwide, not just for electricity but also for heating and transport, every nation would need them and we would need 20,000 nuclear reactors. That's six times more nuclear nations and fifty times more nuclear power plants. (Abbott, 2012) The probability of proliferation rises accordingly. More if you take reprocessing into account.

One path to bomb-making is reprocessing spent fuel from civilian power plants. Reprocessing extracts up to 70 times more energy from the original uranium. (WNA, 2024) It is clearly the way to go if we want to scale up nuclear. Unfortunately, reprocessed fuel is also ideal input for weapons. India got its atom bomb from a Canadian reactor design that had provided the material for the Manhattan bomb building program, combined with a US reprocessing plant. Both under the "Atoms for Peace" program. (Glaser, 2007) (Nuclear Museum, n.d.)

Then there is thorium. Thorium advocates often claim the technology is "proliferation proof," but this requires ignoring the well known protactinium loophole. I've looked at that loophole in some detail and I am comfortable claiming that it actually makes bomb-making with thorium *easier* than with uranium. (Ashley, 2012) (Kang, 2001) Five kilograms of bombgrade U-233, enough for one Hiroshima, (Alvarez, 2013) (Cochran, 1998) can be made from just one tonne of lightly regulated thorium that fits in the back of any van. All you need on top of that is a small research reactor, some chemical separation equipment, and one month of patience. (Ashley, 2012)

Without reprocessing, running the world on nuclear means running out of uranium within five years: a non-solution. (Abbott, 2011) (There is discussion about this, but even if it was a bit longer, it would still not be a solution.) And even "regular" reactors that don't use reprocessing need enriched uranium. You enrich uranium using centrifuges, and as Iran has demonstrated, simply running them a bit longer produces weapons-grade fuel.

A Pakistani scientist known as A. Q. Khan ran a black market for nuclear technology that sold to Libya, Iran, and North Korea for decades before being discovered. (Corera, 2006) For some of the material we still don't know where it ended up. For terrorists with less contacts or funds there is the dirty bomb. Or you can create a crisis using an existing nuclear power plant as Russia did at Zaporizhzhia. Or you could use explosives on a lightly defended spent-fuel pool.

Pretending that we can scale up nuclear without more actors acquiring nuclear weapons requires a faith in international institutions and human restraint that history does not support.

3.5 Nuclear Is a Bad Fit for a Decentralized Grid

Nuclear can scale up and down, (Jenkins, 2018) but in order to keep the already high cost within limits, nuclear plants must produce a high output all day, all year. But a future grid with millions of rooftop solar systems, neighbourhood wind turbines, batteries, electric vehicles, and local energy communities behaves differently: power appearing and disappearing in response to sunlight, wind, and local demand.

Solar and wind have zero marginal cost. (Sensfuss, 2008) Once built, the electricity is essentially free whenever the sun shines or the wind blows. That means nuclear often sits idle while free electrons flow from renewables. And for nuclear, every hour of idleness means the enormous investment must be paid back over fewer kilowatt-hours produced. A huge battery could solve that problem, but would drive up the price further.

Our grid is evolving toward local systems that stabilize themselves at the house, neighbourhood, and region level to avoid congestion. I call them holons. Large centralized plants are a poor fit for this new system. They represent concentrated supply trying to serve distributed demand, with all the fragility and infrastructure costs that entails.

Nuclear is not a villain. But it was built for a world that is fading. In a centralized, no-renewables universe, I would be enthusiastically pro-nuclear. We do not live in that universe.

3.6 The Verdict

I want to be clear about what I am not saying. I am not saying nuclear is evil, or that we should be afraid of accidents, or that existing plants should be shut down prematurely, or that the people who work in the nuclear industry are doing something wrong. I myself consider nuclear better than coal on almost all metrics, and many nuclear advocates are thoughtful people who have arrived at different conclusions through careful reasoning. This is a domain where reasonable people can disagree.

But when I look at the four problems together, losing market share, too expensive to matter, proliferation risks, poor fit for the emerging grid, I see a technology from the age of centralization, trying and failing to compete in the age of distributed energy. I see old thinking: ultimately just another way to produce heat, similar to burning biomass. And I see an opportunity cost, because every euro and every year spent on nuclear is a euro and a year not spent on the technologies that are actually winning.

The dream was beautiful. The reality did not match. And so I was forced to move on.